The supremum side of Erdős problem #1038: per-degree certificates for the $2\sqrt{2}$ bound, and the thirty published decimals of the infimum verified

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October 2026 — draft v0.1 for review; every number interpolated from the record; machine-derived, not peer-reviewed; author list to be agreed

Abstract

Erdős, Herzog and Piranian asked for the infimum and the supremum of the Lebesgue measure of $\{x \in \mathbb{R} : |f(x)| < 1\}$ over monic polynomials f with all roots in $[-1, 1]$. The supremum is $2\sqrt{2}$, attained by $x^2 - 1$; in December 2025 Tao reformulated the problem over probability measures on $[-1, 1]$ and proved the supremum there, with the two-atom measure as the only case of equality. Weights with denominator N are exactly the root measures of degree-N polynomials, so the statement has a decidable instance at every degree. This paper records such decisions, made in exact arithmetic and sharing nothing with the potential-theoretic proof: for degrees $3, 5, 7$ every polynomial has sublevel measure below 2.82, a margin of 0.0084 under $2\sqrt{2}$ that the theorem does not state; for degrees $4, 6, 8$ the degree supremum is placed in $[2\sqrt{2}, 2.82845]$ with the witness $(x^2 - 1)^{N/2}$ at the left end, an enclosure weaker than the theorem by 2.29×10^{-5}; degree 9 exhausted its box budget and is recorded as refused. The certificates are branch-and-bound trees over root boxes, 66,676 boxes in all, pruned by a product-of-distances bound whose own sublevel measure is computed by the same certified Sturm machinery. Certified landscape data show the cubic maximum at an interior root, $r = 201/256$, and both families losing an interior interval of their sublevel set past their champion. Separately, the thirty decimals of the infimum constant $D = 1.83443047576266171109075363512 5\ldots$ printed in the July 2026 manuscript of Darvas, Peng and Tao are decided against an enclosure of width below 10^{-40} obtained by a Krawczyk argument: every printed constant is correct, and the thirtieth decimal is a half-up rounding of an expansion that continues $\ldots 47861\ldots$.

1 Introduction

The problem. For a non-constant monic polynomial $f \in \mathbb{R}[x]$ with all roots in $[-1, 1]$, let $E_f = \{x \in \mathbb{R} : |f(x)| < 1\}$. Erdős, Herzog and Piranian [1, p. 131] asked for the infimum and the supremum of $|E_f|$; it is Problem #1038 of the Erdős problems database [6]. They proved $|E_f| \le 2\sqrt{2}$ when every root lies in $\{-1, 1\}$ and conjectured this bound to be best possible in

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general, and noted that the infimum is below 2 (the example $(x+1)(x-1)^m$, $m \geq 3$) and that with roots in $[-2,2]$ it is 0 [6]; the polynomial lower bound they conjectured for the $[-2,2]$ variant was proved by Pommerenke [2]. For the supremum, the account in [11] attributes the value $2\sqrt{2}$ to Elbert [3, 4] and records Erdős's later request for a more elementary proof [5].

The measure formulation. On 16 December 2025 Tao opened a GitHub issue on the database [7] restating the problem over discrete probability measures $\mu = \sum_i p_i \delta_{a_i}$ on $[-1,1]$ with logarithmic potential $U_\mu(x) = \int \log|x-y|\, d\mu(y)$, asking for the infimum and the supremum of $|\{x : U_\mu(x) < 0\}|$, and wrote that the supremum "conjecturally is attained at $2\sqrt{2}$" by the uniform measure on $\{-1,+1\}$ and that this "may be hard to prove completely." Within a week the problem's forum thread had a proof: on 21 December 2025 Tao posted a write-up completing it, with a dual measure in the intermediate regime suggested by AlphaEvolve, and two participants recorded the supremum problem as finished the same day [8]. The notes of 27 December 2025 [9] state the result as their Theorem 2.1: the supremum of $L(\mu) = |\{x : U_\mu(x) > 0\}|$ (the notes take the potential with the opposite sign) over probability measures on $[t_0 - 1, t_0 + 1]$ is $2\sqrt{2}$, and $L(\mu) \geq 2\sqrt{2}$ forces $\mu = \frac{1}{2}\delta_{t_0-1} + \frac{1}{2}\delta_{t_0+1}$. The problem page has recorded $\sup = 2\sqrt{2}$ since, together with $1.519 \leq \inf \leq 1.835\cdots$ [6].

The infimum. The infimum is the subject of three proof claims on the database's claims page, all AI-assisted and all reporting the same constant: by Wang (14 July 2026), by Darvas, Peng and Tao (15 July 2026) and by Budala (24 August 2026); the page states that "appearing on this page is no guarantee of proof correctness, and does not mean that anyone associated with this site has examined any part of the proof" [10]. The manuscript of Darvas, Peng and Tao [11] defines its constant D from the unique zero of a three-dimensional nonlinear system and prints $D = 1.83443047576266171109075363512 5\ldots$ to thirty decimals. The companion paper [13] brackets the infimum unconditionally, $1.828 \leq \inf \leq 1.8344304971959906$, assuming none of the three claims; its upper end sits 2.14×10^{-8}above D.

What this paper records, and a correction of framing. The campaign whose record this paper interpolates [14] was run on 2026-09-01, framed on the GitHub issue of 16 December 2025, and its report page describes the supremum as Tao's conjecture. The forum resolution of 21 December 2025 and the notes [9] predate the campaign, so the per-degree statements below are not progress on an open conjecture. They are decisions of finite-degree instances of a theorem of the literature, made in exact rational arithmetic by an instrument that shares no idea and no code with the potential-theoretic proof, and they are reported in a fixed grammar: a verdict, a witness, and refusal as a result in its own right. Three things in the record are not given by the theorem: an explicit margin at the odd degrees ($\sup_N < 2.82$, that is 0.0084 below $2\sqrt{2}$, for $N = 3, 5, 7$), certified landscape data for the families that approach the witness, and a refusal at degree 9 with the budget that produced it. The second half of the paper is unrelated to the supremum: it verifies the printed decimals of the infimum constant of [11] against a certified enclosure, in a verifier that executes no line of the authors' code.

Conventions. We follow the sign of [7]: $U_\mu(x) = \int \log|x-y|\, d\mu(y)$ and the set of interest is $\{U_\mu < 0\}$. For $\mu = \sum_i (k_i/N)\delta_{a_i}$ with positive integers k_i summing to N,

$$U_\mu(x) = \frac{1}{N}\log|q(x)|, \qquad q(x) = \prod_i (x - a_i)^{k_i},$$

so $\{U_\mu < 0\} = E_q$ for a monic polynomial of degree N with all roots in $[-1,1]$, and conversely every such polynomial arises. Write $\sup_N$ for the supremum of $|E_q|$ over monic q of degree exactly N with all roots in $[-1,1]$. A statement is *certified*, in this paper, when it is decided by exact rational arithmetic or by outward-rounded interval arithmetic with the certificate on disk in the repository; everything else is called decided, proved or recorded.

2 Statement of results

Theorem 1 (odd degrees)**.** *For $N \in \{3,5,7\}$, every monic polynomial of degree N with all roots in $[-1,1]$ satisfies $|E_q| < 141/50 = 2.82$. In particular $\sup_N \le 2.82 < 2\sqrt{2}$, a margin of 0.0084. The certificates are branch-and-bound trees of 127, 1,583 and 28,773 boxes respectively, each leaf carrying a certified upper bound below the threshold.*

Theorem 2 (even degrees)**.** *For $N \in \{4,6,8\}$, every monic polynomial of degree N with all roots in $[-1,1]$ satisfies $|E_q| < 56569/20000 = 2.82845$, and $q = (x^2-1)^{N/2}$ has $|E_q| = 2\sqrt{2}$. Hence $\sup_N \in [2\sqrt{2}, 2.82845]$. The certificates have 549, 4,219 and 31,425 boxes.*

The left end of Theorem 2 is elementary: $|(x^2-1)^{N/2}| < 1$ exactly when $|x^2-1| < 1$, and $\{|x^2-1| < 1\} = (-\sqrt{2},\sqrt{2}) \setminus \{0\}$. The upper end is weaker than Theorem 2.1 of [9], which gives $\sup_N = 2\sqrt{2}$ for every even N, by 2.29×10^{-5}; the two statements are reached by unrelated routes, and the certificate cannot be tightened to $2\sqrt{2}$ itself because the witness attains it (Section 3.3).

Lemma 1 (continuity in the roots)**.** *For fixed N, the map $r \mapsto |E_{q_r}|$, $q_r(x) = \prod_{i=1}^N (x - r_i)$, is continuous on $[-1,1]^N$. Consequently $\sup_N$ is attained.*

Proof. If $|x| \ge 3$ and $r \in [-1,1]^N$ then every factor has $|x - r_i| \ge 2$, so $|q_r(x)| \ge 2^N > 1$; thus $E_{q_r} \subseteq (-3,3)$ for all r. Fix r^0. For x with $|q_{r^0}(x)| \ne 1$ the indicator of E_{q_r} at x is constant for r near r^0, by continuity of $(x,r) \mapsto q_r(x)$; the exceptional set $\{x : |q_{r^0}(x)| = 1\}$ is finite, being the zero set of the two nonzero polynomials $q_{r^0} \mp 1$. Dominated convergence on $(-3,3)$ gives $|E_{q_r}| \to |E_{q_{r^0}}|$. A continuous function on the compact set $[-1,1]^N$ attains its supremum. □

Corollary 1. *For every odd N, $\sup_N < 2\sqrt{2}$.*

Proof. By Lemma 1 the supremum is attained by some q, whose root measure μ has weights in $\frac{1}{N}\mathbb{Z}$. If $|E_q| \ge 2\sqrt{2}$ then, by the equality clause of Theorem 2.1 of [9] with $t_0 = 0$, $\mu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$, whose weights are not in $\frac{1}{N}\mathbb{Z}$ for odd N. □

Corollary 1 is a consequence of Tao's theorem, not of the certificates; what Theorem 1 adds for $N = 3,5,7$ is the explicit margin. We have not found a quantitative margin for odd degrees stated elsewhere and make no claim beyond the record.

Proposition 1 (certified landscape)**.** *The following sublevel measures are certified enclosures (Table 1, Table 4, Figure 1).*

1. *The witness $x^2 - 1$ has $|E_q| \in [2.828427124745,\ 2.828427124747]$, an enclosure of width 1.1×10^{-12} containing $2\sqrt{2}$, as does $(x^2-1)^2$; x, x^2 and $(x-1)^2$ enclose 2.*
2. *The cubic $(x^2-1)(x-r)$ at $r = 201/256$ has $|E_q| \in [2.754173,\ 2.754174]$, the largest value in the record for degree 3, at an interior root rather than at $r \in \{0, \pm 1\}$: the 1/8-grid sweep and the certified curve on $[1/2, 1]$ both fall away from it. Hence $\sup_3 \in [2.754173, 2.82)$.*

3. *The quintic* $(x^2-1)^2(x-r)$ *at* $r = 905/1024$ *has* $|E_q| \in [2.801090,\ 2.801091]$*, the largest value in the record for degree* 5*. Hence* $\sup_5 \in [2.801090, 2.82)$*.*

4. *Past* $r = 905/1024$ *the sublevel set of the quintic family is two intervals: at* $r = 3622/4096$ *a gap* $[-0.2177, -0.2153]$ *has opened and the measure is* 2.798779*; at* $r = 3664/4096$ *the gap is* 0.117 *wide and the measure is* 2.686190*. The cubic family splits in the same way past* $r = 201/256$*: at* $r = 101/128$ *its sublevel set is two intervals with measure* 2.726610*.*

Proposition 2 (the thirty decimals, decided)**.** *Let* (q^*, u^*, v^*) *be the zero of the system* $F = (F_R, F_L, K)$ *of Appendix A of [11] and* $D = r^* - z^*$ *its Definition A.2. A verifier that re-derives* F *from the displayed equations (A.1)–(A.8) alone, over two interval backends of this repository, decides the following. The zero exists and is unique in a box containing the manuscript's box* B_* *of radius* 10^{-30} *(interval Krawczyk operator, contraction in* 1 *round, radius* 3.84×10^{-14}*), hence the manuscript's triple and the verifier's are the same zero and Lemma A.1 of [11] holds; a fixed-point refinement encloses each coordinate to width below* 3.18×10^{-45}*; and every decimal expansion the manuscript prints for* $q^*, u^*, v^*, p^*, \alpha, r^*, z^*$ *and* D *(Table 6) agrees with the certified enclosure, nine of them as truncations and the thirty-decimal* D *of its page 2 as a half-up rounding: the expansion is*

$$D = 1.83443047576266171109075363512478614302 08\ldots,$$

so its thirtieth decimal is 4 *where the manuscript prints* 5*. Four planted corruptions of the claim are rejected by the same arithmetic.*

Proposition 2 concerns the definition of D and its printed digits. It says nothing about the extremality of D, which is the content of the manuscript's main theorem and was not audited (Section 6).

3 The instrument

3.1 The sublevel measure of one polynomial

Roots are given as fractions n_i/d over one common denominator d, with multiplicities m_i and $|n_i| \le d$. Then $d^N q(x) = A(x) := \prod_i (dx - n_i)^{m_i}$ has integer coefficients, and the boundary of E_q consists of real roots of the integer polynomials $A - d^N$ and $A + d^N$. The instrument isolates the real roots of the squarefree parts of both in $(-3, 3)$ by Sturm sequences over $\mathbb{Z}$ (the implementation is the one calibrated on Mercer's closed forms in the repository's $\lambda(4)$ campaign; for the classical algorithm see [17]), refines each root to a rational enclosure of width 2^{-40}, and refuses if two enclosures overlap, since overlapping enclosures would scramble the gap structure. Each gap between consecutive boundary roots is decided by one exact comparison $|A(t)| < d^N$ at a rational test point t, in $\mathbb{Z}$ after clearing denominators; the first and last gaps must be outside, or the instrument refuses. The measure is then summed with outward rounding: a maximal run of inside gaps from the i-th to the j-th boundary root contributes a length in $[\ell_j - u_i,\ u_j - \ell_i]$, where $[\ell, u]$ are the root enclosures. The result is a true enclosure [lo, hi] of $|E_q|$; no floating-point number enters any decision. The witness enclosure of Proposition 1 is [7774721279937/2748779069440, 388736063997/137438953472] as exact rationals.

3.2 A certified upper bound over a box of root configurations

Let $B = I_1 \times \cdots \times I_N$ be a box of root intervals $I_i = [\ell_i, u_i] \subseteq [-1, 1]$ and, for $x \in \mathbb{R}$, $g_B(x) = \prod_i \operatorname{dist}(x, I_i)$.

Lemma 2. *For every $r \in B$, $E_{q_r} \subseteq \{x : g_B(x) < 1\}$; hence $\sup_{r \in B} |E_{q_r}| \le U(B) := |\{g_B < 1\}|$.*

Proof. $|x - r_i| \ge \operatorname{dist}(x, I_i)$ for $r_i \in I_i$, so $|q_r(x)| \ge g_B(x)$ pointwise. □

g_B vanishes on every I_i and, on each maximal region between or outside the interval endpoints, equals $\prod_{I_i \text{ left of the region}}(x - u_i) \prod_{I_i \text{ right}}(\ell_i - x)$, a polynomial with rational coefficients. $U(B)$ is therefore computed by the machinery of Section 3.1 region by region (Sturm isolation of $g_B = 1$, one exact comparison per sub-gap) and the pieces are merged as an outer measure; the branch-and-bound uses its upper endpoint, so every pruning decision is conservative. On a thin box ($\ell_i = u_i$) the bound is the measure itself; the battery checks that identity at every run on the cubic champion.

3.3 Branch and bound, and the three verdicts

`bnb`(N, T) subdivides the ordered boxes $\{r_1 \le \cdots \le r_N\} \subseteq [-1, 1]^N$, with the mirror $r \mapsto -r$ (which preserves the measure) used to skip boxes whose configurations all have negative sum, splitting the widest coordinate, until every leaf has $U(B) < T$. If the recursion terminates, Lemma 2 gives $|E_{q_r}| < T$ for every $r \in [-1, 1]^N$, collisions and multiplicities included, and the certificate is the tree of leaves. It terminates in one of three ways: every leaf certified (the theorem); the depth cap hit, in which case the instrument reports the box and its bound; or the box budget exhausted. The last two are refusals and are recorded as such; a degree marked refused was attempted and is open, not dropped. For even N the witness attains $2\sqrt{2}$, so no threshold $T \le 2\sqrt{2}$ can ever be certified: the branch-and-bound would recurse forever on the boxes containing $(-1, \ldots, -1, 1, \ldots, 1)$. The campaign therefore takes $T = 56569/20000$, just above $2\sqrt{2}$, for even degrees and $T = 141/50$ for odd ones; the thresholds are exact rationals and are compared with a certified enclosure of $2\sqrt{2}$ before any run.

3.4 Calibrations and the battery

Table 1 lists the calibrations the page and this paper recompute at every build. The battery of the instrument (`instruments/sublevel/battery.js`, 6 checks, 3 falsifiers, run live for this document in 75 ms) re-proves the degree-3 theorem and the degree-4 localisation from scratch at every run and holds three controls that must fire: a threshold below a certified witness must be refused, never certified; a root outside $[-1, 1]$ must be rejected; and a refinement width too coarse to separate close boundary roots must trip the overlap guard rather than merge two roots silently.

4 Results

4.1 The theorem ladder

Table 2 is the record of 2026-09-01 (`certs/sublevel-tao179.json`, repository commit `cd057d7`; the whole campaign took 48.1 minutes, of which the six certificates took 15 s and the refused

polynomial	certified measure of E_q	exact value	boundary roots
x^2-1	$[2.828427124745,\ 2.828427124747]$	$2\sqrt{2}$	3
x^2	$[1.999999999999,\ 2.000000000001]$	2	2
x	$[1.999999999999,\ 2.000000000001]$	2	2
$(x-1)^2$	$[1.999999999999,\ 2.000000000001]$	2	2
$(x^2-1)^2$	$[2.828427124745,\ 2.828427124747]$	$2\sqrt{2}$	3

Table 1: Calibrations, re-certified for this document. The last row is why even degrees differ from odd ones: a power of the witness keeps its sublevel set.

N	certified	T	boxes	leaves	depth	s	
3	$\sup_3 < 2.82$	141/50	127	45	16	0.01	strict
4	$\sup_4 \in [2\sqrt{2},\ 2.82845]$	56569/20000	549	213	59	0.06	localized
5	$\sup_5 < 2.82$	141/50	1,583	522	37	0.21	strict
6	$\sup_6 \in [2\sqrt{2},\ 2.82845]$	56569/20000	4,219	1,370	89	0.65	localized
7	$\sup_7 < 2.82$	141/50	28,773	9,472	63	8.8	strict
8	$\sup_8 \in [2\sqrt{2},\ 2.82845]$	56569/20000	31,425	9,330	118	5.7	localized

Table 2: The per-degree certificates. *boxes* counts every box explored, *leaves* the boxes closed by $U(B) < T$, *depth* the deepest split. Degree 9 is not in the table: it is refused (`bnb: box budget exhausted`) after 47.9 minutes within a budget of 1,500,000 boxes and depth 160.

degree 47.9 minutes). Every box count, leaf count and depth is read from the record; the battery's live re-proofs of degrees 3 and 4 reproduce their box counts exactly.

In measure language, Table 2 says: every discrete probability measure on $[-1,1]$ whose weights have denominator 3, 5 or 7 has $|\{U_\mu < 0\}| < 2.82$, and every one whose weights have denominator 4, 6 or 8 has $|\{U_\mu < 0\}| < 2.82845$, the two-atom measure attaining $2\sqrt{2}$. The budgets were 500,000 to 3,000,000 boxes and depths 40 to 200 (Table 3); every certificate fits its budget by a wide factor, and degree 9 did not.

N	box budget	depth cap
3	500,000	40
5	3,000,000	70
7	5,000,000	100
4	3,000,000	120
6	3,000,000	160
8	1,500,000	200
9	1,500,000	160

Table 3: The budgets each degree was given, read from the campaign source.

4.2 The landscape

Before any theorem was attempted the campaign swept root grids for certified lower-bound records (Table 4, 838 configurations), then refined the two shapes that stood out. The degree-3 champion is at an interior root: on the grid of step 1/128 the family $(x^2-1)(x-r)$ peaks at $r = 100/128$ with 2.753089, and the record's refinement $r = 201/256$ certifies $[2.754173,\ 2.754174]$.

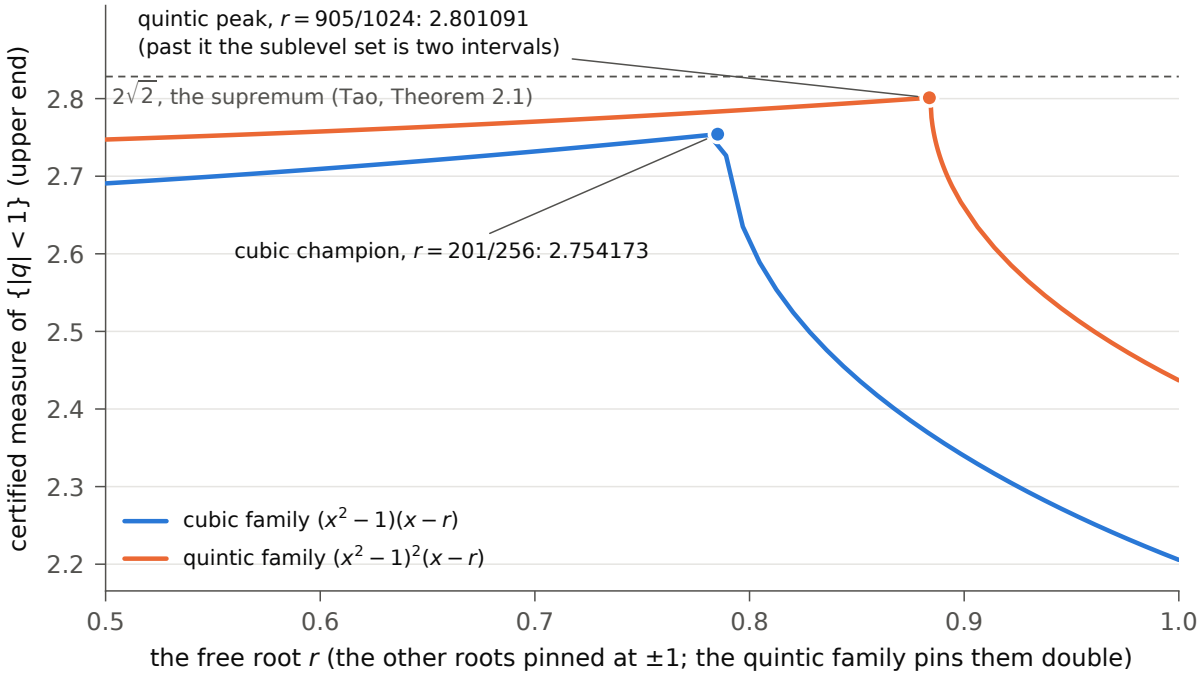


Figure 1: The two families against the free root r: 163 certified enclosures recomputed for this document (the plotted value is the upper end; widths are far below the resolution of the picture). The quintic family is drawn at step 1/128 and, between $r = 3600/4096$ and $3664/4096$, at step 1/4096. Both families reach their maximum where their sublevel set is still one interval and descend as soon as it splits.

The family $(x^2-1)^2(x-r)$ climbs higher, to $[2.801090,\ 2.801091]$ at $r = 905/1024$, and both families then descend steeply (Figure 1): the next grid point past the cubic champion, $r = 101/128$, already measures 2.726610, and the quintic family loses 0.002 within 1/4096 of its peak.

N	grid d	configurations	champion roots	certified measure
3	8	505	$(-1,\ 3/4,\ 1)$	$[2.744632,\ 2.744633]$
4	4	264	$(-1,\ -1,\ 1,\ 1)$	$[2.828427,\ 2.828428]$
5	2	69	$(-1,\ -1,\ 1/2,\ 1,\ 1)$	$[2.747268,\ 2.747269]$

Table 4: The grid sweeps of the record: all root configurations n_i/d with $|n_i| \le d$ and non-negative sum, each certified, and the champion of each sweep.

The descent is steep but continuous, as Lemma 1 requires; the report page's description of it as a discontinuous drop is corrected here. The mechanism is visible in the certified run structure: up to $r = 3620/4096$ the quintic sublevel set is one interval and the measure is 2.801091; at $r = 3622/4096$ a gap $[-0.2177, -0.2153]$ of width 0.002 has opened inside it and the measure is 2.798779; by $r = 3664/4096$ the gap is 0.117 wide. A local maximum of $|q_r|$ inside the interval has crossed the level 1; when such a maximum is nondegenerate the excluded interval has width of order $(r - r^*)^{1/2}$, which is the shape of the curve past each champion. At

$r = 1$ the cubic family measures 2.205570 and the quintic 2.436924.

5 The thirty decimals of the infimum constant

5.1 What is defined, and what is checked

Appendix A of [11] defines, with $\Lambda(y) = \log \frac{1+y}{1-y}$, $p(q) = \log \frac{2}{1-q^2}/\Lambda(q)$ and $X_q(y) = \frac{2q^2-1-y^2}{1-y^2}$, the functions $F_R(q,u) = \Lambda(u) - p(q)\log\frac{q+u}{q-u}$, $F_L(q,v) = \Lambda(v) - p(q)\log\frac{q+v}{v-q}$ and a third equation $K(q,u,v)$ built from them and from the displayed derivatives (A.4)–(A.6); the extremal triple is the zero of $F = (F_R, F_L, K)$ inside the box B_* of radius 10^{-30} about centres (q_0, u_0, v_0) printed to 35 decimals (Table 5), and $D = r^* - z^*$ with $r^* = X_{q^*}(u^*)$, $z^* = X_{q^*}(v^*)$. The manuscript proves existence by a scalar intermediate-value argument on faces of B_* and ships Arb and SymPy certificates of its own [11].

centre	printed in (A.9)	certified expansion
q_0	0.94985834582347031848349802544031979	0.94985834582347031848349802544031979…
u_0	0.89396612207545611611103128550805002	0.89396612207545611611103128550805002…
v_0	0.96455509699582140639698562612728156	0.96455509699582140639698562612728156…

Table 5: The box centres of the manuscript against the certified enclosures, which are thin to 40 decimals; the printed centres are the first 35 decimals of the zero.

The verifier, `verify.js` (its path is given in the Reproduction section), was written against the manuscript on 4 August 2026 and filed on the authors' repository the next day [12]; it is kept byte-identical and re-run, staged against this repository's interval toolkit, at every build of the report page and for this document (26 checks, 0 failed, 4 mutation controls rejected, 57 ms). It executes no line of the authors' code and takes no number from it except the claims under audit and the centres of (A.9), which are part of the claim. One implementation of F with forward-mode automatic differentiation runs over two backends: outward-rounded IEEE-754 double intervals with the interval logarithm of the `eqcert` toolkit [16], and a fixed-point BigInt interval layer at quantum 10^{-48} with directed rounding, the logarithm by the atanh series after an exact power-of-two normalisation with an explicit tail bound. The checks, in order:

1. the automatic derivatives overlap the manuscript's displayed closed forms (A.4)–(A.6);
2. the face inequalities (A.12)–(A.14) hold on $I_* = q_0 \pm 10^{-31}$ and the u, v faces of B_*, in a first-order mean-value form $F(m) + J(B)(B - m)$ — the F_R faces evaluate to an interval between 4.05×10^{-30} and 6.42×10^{-30} and to its negative, the F_L faces to intervals between 7.84×10^{-29} and 9.04×10^{-29} in absolute value, so close to zero that naive interval evaluation over the printed radii is too wide to decide the sign;
3. the interval Krawczyk operator [15] contracts in 1 round with radius 3.84×10^{-14}, proving existence and local uniqueness of the zero; this is a different route from the manuscript's, and B_* lies inside the uniqueness box, so the two triples are the same zero;
4. iterating $X \leftarrow K(X)\cap X$ in the fixed-point layer refines the enclosure to widths 1.64×10^{-45}, 3.18×10^{-45}, 1.12×10^{-45}in q, u, v, with the first high-precision image re-checked to land in the interior; the refined q^* lies in I_* and the triple in B_*, which is Lemma A.1 of [11];

5. the derived constants are enclosed and every printed expansion is compared digit for digit (Table 6): a printed string matches as a *truncation* if the enclosure pins the truncation at that many decimals and it equals the string, as a *rounding* if the same holds for the half-up rounding;

6. four mutation controls must be rejected: the last printed digit of D bumped ($\ldots 125 \to \ldots 126$); the sign inside K flipped, after which F_3 at the certified triple is 1.307, not zero; the box shifted by 10^{-10} in q, where F_R is 1.19×10^{-9}away from zero; and the constant 2 inside $p(q)$ perturbed to 2.000000001, which moves F_R at the triple to -4.78×10^{-10}.

constant	page	printed in [11]	certified expansion	matches as
D	p.2	1.83443047576266171109075363512 5	1.834430475762661711090753635124786…	rounded
α	p.5	0.804461754260998666	0.80446175426099866684 1…	truncation
q^*	p.24	0.949858345823470318	0.94985834582347031848 3…	truncation
u^*	p.24	0.893966122075456116	0.89396612207545611611 1…	truncation
v^*	p.24	0.964555096995821406	0.96455509699582140639 6…	truncation
p^*	p.24	0.824521729252520379	0.82452172925252037928 5…	truncation
α	p.24	0.804461754260998666	0.80446175426099866684 1…	truncation
r^*	p.24	0.0263231076477328506	0.026323107647732850635 1…	truncation
z^*	p.24	-1.808107368114928860	$-1.808107368114928860455\ldots$	truncation
D	p.24	1.834430475762661711	1.834430475762661711090…	truncation

Table 6: Every decimal expansion the manuscript prints for its extremal constants, against the certified enclosures (thin to 40 decimals; shown to three decimals past the printed length). Nine match as truncations; the thirty-decimal headline D matches as a half-up rounding.

5.2 The thirtieth decimal

The headline value on page 2 of [11] is correct to under 5×10^{-31}: it is the half-up rounding at thirty decimals of the certified expansion,

$$\begin{aligned} \text{printed (page 2):} \quad & 1.83443047576266171109075363512 5\ldots \\ \text{certified:} \quad & 1.83443047576266171109075363512\,47861\ldots \end{aligned}$$

Read as an expansion prefix, which a trailing ellipsis conventionally promises, its last digit differs from the expansion's (5 printed, 4 in the expansion). This is a presentation detail, not an error in the mathematics, and it is the kind of detail only digit-level certified arithmetic can see; the same half-unit class is the benign end of the failure taxonomy of the repository's audit of Erdős #852 [14]. The enclosures sit more than 14 decades inside the box the manuscript's Lemma A.1 asks for, which is what makes the printed decimals decidable rather than merely consistent.

6 What is not claimed

- *The supremum theorem is not ours.* $\sup_\mu |\{U_\mu < 0\}| = 2\sqrt{2}$ over all probability measures on $[-1, 1]$, with the two-atom measure as the only case of equality, is Theorem 2.1 of [9]; for polynomials the value is attributed to [3, 4] in [11]. The certificates of Theorems 1 and 2 neither use nor replace it, and for even degrees their upper end is weaker than it. Degrees 1 and 2 are not in the record (degree 1 is the trivial measure 2).

- *Degrees beyond* 8 *are not decided.* Degree 9 is refused on its budget; nothing is asserted about it or about any higher degree. The ladder is compute-bound, not idea-bound, but that is a conjecture about the instrument and is not certified.
- *The equality case is not characterised by the certificates.* That the witness is the only maximiser at even degrees follows from [9], not from the localisation.
- *The landscape is data.* The cubic champion and the quintic peak are certified values of explicit polynomials, hence certified lower bounds on $\sup_3$ and $\sup_5$; that they are the maxima of their families is not certified (the grids are finite), and no lower witness for degree 7 is in the record.
- *The infimum is not decided here.* Proposition 2 verifies the definition of D and its printed digits. The extremality proof of [11], the other two proof claims, the manuscript's own Arb and SymPy certificates (read for scope only) and its AI-use chronology were not audited. The unconditional bracket on the infimum is the companion paper's [13].
- *Nothing here is peer-reviewed.* The results are machine-derived; the repository re-derives every number at every build, and refutations are invited.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records, the instruments and the commands, in the repository [14] at commit `16849460e608`:

- `node tools/run-sublevel-campaign.js`
  writes the supremum record `certs/sublevel-tao179.json` (about 48.1 minutes, most of it the refused degree); the instrument is `instruments/sublevel/` (`measure.js`, `bound.js`, `battery.js`).
- `node instruments/sublevel/battery.js`
  re-proves degrees 3 and 4 and fires the three controls in well under a second.
- `node tools/build-report-erdos1038-sup.js`
  rebuilds the report page (`https://carlostoledo.co/reports/erdos1038-sup.html`), re-certifying the witness, the calibrations and both family curves.
- `node tools/build-report-lemniscate.js`
  re-runs the thirty-decimal verifier, kept byte-identical at `legacy/research/challenges/lane/laneb-lemniscate/verify.js`, against `instruments/interval` and rebuilds the verification page `https://carlostoledo.co/reports/verify-lemniscate.html`; the earlier audit page is `https://carlostoledo.co/reports/claim-lemniscate.html`.
- `node tools/paper-numbers/erdos1038-sup.js`
  writes the macros of this document, the curve points of Figure 1 and the figure itself, running the battery and the verifier live and refusing when a record no longer supports a sentence; `node tools/build-paper-tex.js erdos1038-sup` compiles it.

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